[EDR/XDR Basics - Endpoint & Extended Detection Engineering]

[What is EDR?]

EDR (Endpoint Detection and Response) is a security solution that continuously monitors endpoints (laptops, desktops, servers, and workstations) to detect, investigate, and respond to cyber threats.

[EDR Main Functions]

- **Continuous Monitoring:** Monitors endpoint activities 24/7.
- **Threat Detection:** Detects malware, ransomware, suspicious processes, and attacks.
- **Investigation:** Provides detailed logs and timelines.
- **Response:** Can isolate infected devices, kill processes, and remove threats.
- **Threat Hunting:** Helps SOC analysts search for indicators of compromise (IOCs).

Practical Scenario Example:

> A user downloads a malicious file.
> **Antivirus Approach:** Detects known malware signatures and blocks it.
> **EDR Approach:** Detects the file, monitors its active behavior, sees PowerShell spawning suspicious commands, alerts the SOC team, and isolates the endpoint automatically.

[What is XDR?]

XDR (Extended Detection and Response) extends EDR capabilities by collecting and correlating data from multiple security layers across the corporate infrastructure.

[XDR Telemetry & Data Sources]

```
[ XDR Unified Architecture ]
   ├── Endpoints (EDR System Logs)
   ├── Network Devices & Frameworks
   ├── Firewalls (Boundary Traffic)
   ├── Email Security Gateways
   ├── Cloud Platforms & Storage
   └── Identity Systems (Active Directory / IAM Platforms)
```

[XDR Main Functions]

1. Provides complete centralized visibility across disparate layers.
2. Correlates isolated alerts from multiple sources automatically.
3. Dramatically reduces false positives.
4. Executes automated response playbooks across multiple security layers simultaneously.
5. Accelerates incident investigation speed for analysts.

Correlated Threat Workflow Example:

```
[ Phishing Delivery ] ──> Email Security detects suspicious incoming mail.
                                       │
                                       ▼
[ User Interaction ]  ──> User clicks the link; Endpoint executes malicious payload.
                                       │
                                       ▼
[ Network Connection ] ──> Host contacts external server; Firewall logs bad IP connection.
                                       │
                                       ▼
[ XDR Incident Engine ] ──> Combines ALL events into a single managed incident timeline.
```

[Antivirus vs EDR]

Feature Matrix	Antivirus (AV)	Endpoint Detection & Response (EDR)
Primary Purpose	Prevent known malware installation.	Detect, investigate, and respond to active threats.
Detection Method	Signature-based lookup strings.	Behavioral Analytics + Machine Learning + Signatures.
Monitoring Posture	Limited to execution-time scans.	Continuous 24/7 real-time process monitoring.
Threat Hunting	No structural hunting capabilities.	Yes; raw metadata search capability.
Investigation Data	Limited static logs.	Detailed process trees, timelines, and execution paths.
Response Actions	Quarantine or delete local file.	Isolate network host, kill processes, isolate threads, remediate.
Advanced Attacks	Limited effectiveness.	Strong tracking against fileless and stealth vectors.
SOC Workflow Usage	Basic signature hygiene.	Essential core tool for investigation.

[Detection Coverage Comparison]

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[ Antivirus Limits ]
   ├── Detects ──────> Known Malware, Common Viruses, Static Trojan Signatures
   └── Misses ───────> Fileless attacks, Living-off-the-Land (LotL), APTs, Zero-day attacks

[ EDR Capability ]
   └── Detects ──────> Process creation chains, Network sockets, Encoded execution strings, Per
```

Technical Attack Scenario: An attacker runs an obfuscated line like: `powershell.exe -enc <base64_command>`. An Antivirus might miss it if the payload's hash signature hasn't been blacklisted yet. An EDR identifies the behavior (suspicious argument flags, encoded lines, parent-child process chains) and triggers a SOC alert, kills the running process, and isolates the host automatically.

[Common Industry EDR / XDR Tools]

EDR Solutions Matrix	XDR Extended Solutions Matrix
• CrowdStrike Falcon Insight • Microsoft Defender for Endpoint (MDE) • SentinelOne Singularity EDR • VMware Carbon Black Host Protection • Trend Micro Vision One Endpoint Suite	• Microsoft Defender XDR Suite • Palo Alto Networks Cortex XDR • Trend Micro Vision One XDR • Cisco XDR Ecosystem

[SOC L1 One-Line Quick Reference]

```
• Antivirus  = Prevention Hygiene
• EDR        = Endpoint Detection + Timeline Investigation + Remediation Response
• XDR        = EDR + Network + Email + Cloud + Identity Unified Behavioral Visibility
```